

```
package ict.servlet;
```

```
@WebServlet(name = "YourServlet", urlPatterns = {"/yourServlet"})
```

```
public class YourServlet extends HttpServlet {  
    private YourDB db;  
    @Override  
    public void init() throws ServletException {  
        String dbUrl = getServletContext().getInitParameter("dbUrl");  
        String dbUser = getServletContext().getInitParameter("dbUser");  
        String dbPassword = getServletContext().getInitParameter("dbPassword");  
        db = new YourDB(dbUrl, dbUser, dbPassword);  
    }  
    @Override  
    protected void doGet(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        processRequest(request, response);  
    }  
    @Override  
    protected void doPost(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        processRequest(request, response);  
    }  
    private void processRequest(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        String param = request.getParameter("paramName");  
        if (param == null || param.trim().isEmpty()) {  
            throw new ServletException("Parameter is required.");  
        }  
        try {  
            ArrayList<YourBean> list = db.queryByParam(param);  
            request.setAttribute("list", list);  
            RequestDispatcher rd = request.getRequestDispatcher("/display.jsp");  
            rd.forward(request, response);  
        } catch (Exception ex) {  
            throw new ServletException(ex);  
        }  
    }  
}
```

Servlet

```
package ict.bean;  
public class YourBean {  
    private String id;  
    private String name;  
    private String email;  
    private int age;  
    public YourBean() {}  
    public YourBean(String id, String name, String email, int age) {  
        this.id = id;  
        this.name = name;  
        this.email = email;  
        this.age = age;  
    }  
    // Getter & Setter  
    public String getId() { return id; }  
    public void setId(String id) { this.id = id; }  
}
```

Bean

```
package ict.db;  
public class YourDB {  
    private String dbUrl, dbUser, dbPassword;  
  
    public YourDB(String dbUrl, String dbUser, String dbPassword) {  
        this.dbUrl = dbUrl;  
        this.dbUser = dbUser;  
        this.dbPassword = dbPassword;  
    }  
    try {  
        Class.forName("com.mysql.cj.jdbc.Driver");  
    } catch (ClassNotFoundException e) {  
        throw new RuntimeException("MySQL driver not found", e);  
    }  
    private Connection getConnection() throws SQLException {  
        return DriverManager.getConnection(dbUrl, dbUser, dbPassword);  
    }  
}
```

DB

```
public ArrayList<YourBean> queryAll() throws SQLException {  
    ArrayList<YourBean> list = new ArrayList<>();  
    String sql = "SELECT id, name, email, age FROM  
your_table";  
    try (Connection conn = getConnection();  
        Statement stmt = conn.createStatement();  
        ResultSet rs = stmt.executeQuery(sql)) {  
        while (rs.next()) {  
            list.add(mapRow(rs));  
        }  
    }  
    return list;  
}  
public void addRecord(String id, String name, String email,  
int age) throws SQLException {  
    String sql = "INSERT INTO your_table (id, name, email,  
age) VALUES (?, ?, ?, ?)";  
    try (Connection conn = getConnection();  
        PreparedStatement ps = conn.prepareStatement(sql)) {  
        ps.setString(1, id);  
        ps.setString(2, name);  
        ps.setString(3, email);  
        ps.setInt(4, age);  
        ps.executeUpdate();  
    }  
}  
public void deleteRecord(String id) throws SQLException {  
    String sql = "DELETE FROM your_table WHERE id = ?";  
    try (Connection conn = getConnection();  
        PreparedStatement ps = conn.prepareStatement(sql)) {  
        ps.setString(1, id);  
        ps.executeUpdate();  
    }  
}  
private YourBean mapRow(ResultSet rs) throws  
SQLException {  
    return new YourBean(rs.getString("id"),  
rs.getString("name"),  
rs.getString("email"), rs.getInt("age"));  
}  
}
```

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Input Form</title>
```

```
</head>
```

```
<body>
```

```
<h1>Submit Data</h1>
```

```
<form action="yourServlet" method="post">
```

```
<label>Name:</label>
```

```
<input type="text" name="name" required />
```

```
<br/>
```

```
<label>Email:</label>
```

```
<input type="email" name="email" required />
```

```
<br/>
```

```
<input type="submit" value="Submit" />
```

```
</form>
```

```
</body>
```

```
</html>
```

```
<%@ page import="java.util.ArrayList, ict.bean.YourBean" %>
```

```
<%
```

```
ArrayList<YourBean> list = (ArrayList<YourBean>) request.getAttribute("list");
```

```
%>
```

```
<h1>Results</h1>
```

```
<table border="1">
```

```
<tr>
```

```
<th>ID</th>
```

```
<th>Name</th>
```

```
<th>Email</th>
```

```
<th>Age</th>
```

```
</tr>
```

```
<%
```

```
if (list != null) {
```

```
for (YourBean bean : list) {
```

```
%>
```

```
<tr>
```

```
<td><%= bean.getId() %></td>
```

```
<td><%= bean.getName() %></td>
```

```
<td><%= bean.getEmail() %></td>
```

```
<td><%= bean.getAge() %></td>
```

```
</tr>
```

```
<%
```

```
}
```

```
%>
```

```
</table>
```

web.xml 設定

```
<display-name>YourApp</display-name>
```

```
<!-- Database Parameters -->
```

```
<context-param>
```

```
<param-name>dbUrl</param-name>
```

```
<param-value>jdbc:mysql://localhost:3306/YourDB</param-value>
```

```
</context-param>
```

```
<context-param>
```

```
<param-name>dbUser</param-name>
```

```
<param-value>root</param-value>
```

```
</context-param>
```

```
<context-param>
```

```
<param-name>dbPassword</param-name>
```

```
<param-value></param-value>
```

```
</context-param>
```

```
<context-param>
```

```
<param-name>dbPassword</param-name>
```

```
<param-value></param-value>
```

```
</context-param>
```

```
<!-- Error Page -->
```

```
<error-page>
```

JSP Input

JSP Output

Web.xml

```
<exception-type>java.lang.Exception</exception-type>
```

```
<location>/HandleError.jsp</location>
```

```
</error-page>
```

```
<welcome-file-list>
```

```
<welcome-file>index.jsp</welcome-file>
```

```
</welcome-file-list>
```

```
</web-app>
```

Error page

錯誤處理頁 (Error Handling)

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
```

```
<%@ page isErrorPage="true" %>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Error Page</title>
```

```
</head>
```

```
<body>
```

```
<h1>Ohh! Some error(s).</h1>
```

```
<p><%= (exception != null && exception.getMessage() != null) ? exception.getMessage() : "Invalid input." %></p>
```

```
<a href="index.jsp">Please try again!!!</a>
```

```
</body>
```

```
</html>
```